

Synthesis of Iron-Carbide Nanoparticles: Identification of the Active Phase and Mechanism of Fe-Based Fischer–Tropsch Synthesis

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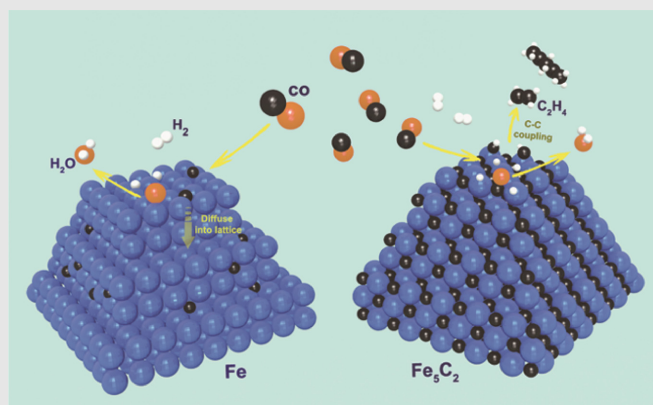
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Despite the extensive study of the Fe-based Fischer–Tropsch synthesis (FTS) over the past 90 years, its active phases and reaction mechanisms are still unclear due to the coexistence of metals, oxides, and carbide phases presented under realistic FTS reaction conditions and the complex reaction network involving CO activation, C–C coupling, and methane formation. To address these issues, we successfully synthesized a range of pure-phase iron and iron-carbide nanoparticles (Fe, Fe₅C₂, Fe₃C, and Fe₇C₃) for the first time. By using them as the ideal model catalysts on high-pressure transient experiments, we identified unambiguously that all the iron carbides are catalytically active in the FTS reaction while Fe₅C₂ is the most active yet stable carbide phase, consistent with density functional theory (DFT) calculation results. The reaction mechanism and kinetics of Fe-based FTS were further explored on the basis of those model catalysts by means of transient high-pressure stepwise temperature-

programmed surface reaction (STPSR) experiments and DFT calculations. Our work provides new insights into the active phase of iron carbides and corresponding FTS reaction mechanism, which is essential for better iron-based catalyst design for FTS reactions.



Keywords: Fischer–Tropsch synthesis, iron carbides, active phase, reaction mechanism

Introduction

Fischer-Tropsch synthesis (FTS) receives increasing attention because synthesis gas (syngas) can be obtained from coal, biomass, and shale gas, which becomes particularly important for the production of alternative fuel and chemicals.^{1,2} Despite the extensive exploration of Fe-based catalysts over the past 90 years, its active phases and reaction mechanisms are still controversial.³ The typical iron catalysts, usually produced from thermal reduction and successive activation of iron oxide precursors, contain different phases, including metals, oxides, and carbides, produced during the pretreatment of the catalysts by carbon-containing gases such as CO.^{4–10} The phase evolution of the iron catalysts during the FTS reaction is even more complicated, and in most cases, a mixture of different phases occurred.^{11–15} To address this issue, various types of metal or metal carbides were prepared.^{16–22} Metallic iron^{23,24} and various phases of iron carbides had all been claimed to be active.^{4,6,25–38} The complication for FTS comes also from the complexity of the reaction network itself, including CO activation, C–C bond formation, and methane formation.^{32,39–45} However, the reaction performance dependence on catalyst phases of iron and carbides as well as the catalytic mechanism behind was still elusive, hindering better Fe-based catalyst design for FTS.

To address these challenging issues, for the first time, we have successfully synthesized a variety of pure-phase iron and iron-carbide catalysts with similar particle size, including Fe, Fe₇C₃, Fe₃C, Fe₅C₂, and so forth, which allow us to identify their initial and intrinsic activities as well as the structural evolution of iron-based catalysts during FTS. At the same time, we developed a new experiment, a transient high-pressure stepwise temperature-programmed surface reaction (STPSR) that enabled us to directly explore such complicated and challenging problems in FTS as fundamental knowledge about syngas activation, hydrocarbon and methane formation on pure-phase catalysts, which were previously unavailable. The comprehensive density functional theory (DFT) calculations revealed a deep insight into the intrinsic activity of iron metal and iron carbides on CO activation, C–C bond, and methane formation, rationalizing the kinetic and thermodynamic origin on the structural evolution of different iron-based catalysts during the FTS reaction. This deeper understanding of the active phase of iron-based catalysts and the corresponding FTS reaction mechanism is beneficial for rational design of more effective Fe-based catalysts on FTS by the synthesis of more Fe₅C₂ catalyst.

Experimental Methods

Synthesis of Fe₇C₃ and Fe₂C NPs

In a four-neck flask, 20 mL of *N,N*-dimethyloctadecylamine (for Fe₇C₃) or dodecylamine

(for Fe₂C) was stirred sufficiently and degassed under 120 °C for 2 h. Then, the system was refilled with NH₃ and heated to 180 °C. After that, Fe(CO)₅ (0.7 mL, 5.0 mmol) was injected under NH₃ atmosphere and kept at this temperature for 30 min. A color change from orange to black was observed during the process, implying the decomposition of Fe(CO)₅ and the nucleation of Fe nanocrystals. Subsequently, the mixture was further heated to 350 °C (for Fe₇C₃) or 260 °C (for Fe₂C) at 10 °C/min and kept for 2 h before it was cooled down to room temperature. The product was washed with ethanol and hexane, and collected for further characterization. The as-synthesized nanoparticles (NPs) were kept in an Ar-filled glovebox to avoid exposure to air before further characterization. In the absence of NH₃, the NPs would be oxidized to iron oxide (Supporting Information Figures S1 and S2).

Synthesis of Fe₅C₂ and Fe NPs

In a four-neck flask, a mixture of octadecylamine (14.5 g) and Hexadecyltrimethylammonium bromide (CTAB) (0.113 g) was stirred sufficiently. Then, the system was refilled with N₂ and heated to 180 °C. Following that, Fe(CO)₅ (0.5 mL, 3.6 mmol) was injected under a N₂ blanket. The mixture was kept at 180 °C for 10 min. A color change from orange to black was observed during the process, implying the decomposition of Fe(CO)₅ and the nucleation of Fe nanocrystals. Subsequently, the mixture was further heated to 350 °C (for Fe₅C₂) or 300 °C (for Fe) at 10 °C/min and kept for 10 min before it was cooled down to room temperature. The product was washed with ethanol and hexane, and was kept in an Ar-filled glovebox to avoid exposure to air before further characterization.

Preparation of supported catalyst

The NPs obtained from high-temperature liquid-phase synthesis had been washed with *n*-hexane and ethanol several times and dispersed in *n*-hexane under N₂ protection. Afterward, the dispersion of iron-carbide NPs was added into a certain amount of silica (N₂ adsorption-desorption isotherm is shown in Supporting Information Figure S3) under stirring. After evaporating the solvent at room temperature, the supported catalyst was prepared. The amount of iron determined by inductively coupled plasma (ICP) was around 8%.

Catalysis reaction

The catalytic performance of the catalysts was evaluated in a fixed bed reactor. About 80 mg catalyst was loaded in a quartz-lined stainless steel reactor. The feed gas was a mixture of 32% CO, 64% H₂, and 4% Ar. In a typical reaction, the pressure and gas flow rate were set for 30 bar and 20 mL/min (gas hourly space velocity,

GHSV = 15,000 mL/(g·h), respectively. Then, the reaction tube was heated from room temperature to 270 °C at 5 °C/min, and the reaction was conducted at 270 °C. It is worth noting that no reduction or carburizing pretreatment was carried out before the reaction. The gas-phase products were analyzed by an Agilent 6890 gas chromatography (GC) equipped with a flame ionization detector (FID) and thermal conductivity detector (TCD), with 4% Ar as inert standard. The heavier hydrocarbons were cooled down and collected in a trap, and analyzed offline by an Agilent GC 7820, with a HP-5 capillary column and FID. The products selectivity was calculated on a carbon basis.

STPSR experiment

Prior to the STPSR experiments, the Fe catalyst was treated in H₂ (20 mL/min) and the Fe₅C₂ catalyst was treated in 10% C₂H₄/H₂ mixture at 300 °C for 2 h to remove the surface contaminants. After cooling down to room temperature, the gas flow was switched to synthesis gas of 20 mL/min, and the pressure was raised to 30 bar. Then, the reactor was heated to 150 °C at 20 °C/min, and kept steady for 2 h. Afterward, the temperature was elevated 20 °C higher and held for 2 h at that temperature. The process was repeated until the reaction temperature reached 270 °C. A Pfeiffer Omnistar mass spectrometer (MS) was used to analyze the reactants and products online. The M/e value detected as follows: 2 for hydrogen; 15 for methane; 18 for water; 26, 27, and 30 for C₂ products (acetylene, ethylene, and ethane); 28 for CO; 42 for C₃; 44 for CO₂; 56 for C₄+, and 70 for C₅+

Catalyst characterization

The transition electron microscopy (TEM) experiments were conducted at a FEI Tecnai F30 transmission electron microscope (HRTEM) operating at 300 keV. The X-ray photoelectron spectroscopy (XPS) experiments were carried out on an AXIS Ultra imaging photoelectron spectrometer with Al K α as X-ray source. The binding energy of graphite carbon was calibrated to 284.8 eV. Powder X-ray diffraction (XRD) data were collected at a Rigaku D/Max-2400 equipped with Cu K α radiation. The Raman characterizations were performed on a Renishaw 1000 Raman imaging microscope system with an excitation wavelength of 632.8 nm. The ⁵⁷Fe Mössbauer effect spectra of as-synthesized iron-carbide NPs were collected by a Topological 500A spectrometer and a proportional counter at room temperature. The γ radiative source was a ⁵⁷Co (Rh) moving with constant acceleration mode. The X-ray absorption fine structure (XAFS) spectra were collected at beam line 14 W of Shanghai Synchrotron Facility (SSRF) in transmission mode with a Si (111) monochromator. The samples for characterization were prepared and transferred under protection of Ar. As metallic iron and iron carbide were very sensitive to oxidation, the oxidation of the sample

was sometimes unavoidable. For TEM study, the catalyst used was passivated in 0.5% O₂ in Ar at room temperature for 1 h before the measurement.

Calculation

Spin-polarized DFT calculations have been performed by using the Vienna ab initio simulation package (VASP).^{46,47} Throughout the calculations, projector augmented wave (PAW) potential⁴⁸ and the generalized gradient approximation (GGA) with the Perdew–Burke–Ernzerhof (PBE) exchange–correlation functional⁴⁹ were adopted. The planewave cutoff energy was set by 400 eV. The force and energy convergence standards were 0.02 eV/Å and 1 $\times$ 10^{−4} eV, respectively. Monkhorst–Pack⁵⁰ *k*-points sampling of 3 $\times$ 7 $\times$ 7 and 10 $\times$ 10 $\times$ 10 were adopted for Fe₅C₂ and Fe bulk calculations with monoclinic (space group C2/c) and body-centered cubic (BCC) crystal structures, respectively. The optimized lattice constant of Fe₅C₂ is *a* = 11.55 Å, *b* = 4.50 Å, and *c* = 4.99 Å with β = 97.6°, which are consistent with experimental findings that *a* = 11.59 Å, *b* = 4.58 Å, and *c* = 5.06 Å with β = 97.75°.⁵¹ The calculated lattice constant (2.83 Å) for Fe with BCC crystal structure also agreed very well with the experiment.⁵²

Fe-terminated Fe₅C₂ (100) surface was simulated by a slab of seven-layered Fe atoms and three-layered C atoms. In the calculations, the topmost four Fe and one C layers were fully relaxed, whereas the remaining atoms were fixed in their bulk positions. For the BCC-Fe (310) surface, a 10-layered Fe atoms slab model was used, and only the top five Fe layers were allowed to relax. A *p*(2 $\times$ 2) unit cell was utilized for the two slab models considered. We have used Monkhorst–Pack mesh *k*-points of 3 $\times$ 3 $\times$ 1 for Fe₅C₂ (100) and 5 $\times$ 5 $\times$ 1 for the BCC-Fe (310) surface. The vacuum region along the *z* direction was specified by 15 Å, and the dipole correction was considered in our calculations. Force reversed method⁵³ was used to locate the transition states (TSs), and a force tolerance of 0.03 eV/Å was applied without zero-point correction. Some TSs along the minimum-energy reaction pathways were also reaffirmed by using the climbing image-nudged elastic band (CI-NEB) method.⁵⁴ For a given elementary reaction, we considered the separately adsorption of intermediates at their most favorable adsorption sites as the initial and final states for the reaction barrier calculations.

Results and Discussion

In fabrication of both Fe₇C₃ and Fe₂C NPs, NH₃ was chosen as the atmosphere as well as the inducing agent while Fe(CO)₅ was used as the precursor. In particular, Fe₇C₃ NPs were obtained in *N,N*-dimethyloctadecylamine solvent under 350 °C for 2 h, and Fe₂C NPs were produced in dodecylamine solvent under 260 °C for 2 h (see Supporting Information). α -Fe

and Fe_5C_2 NPs were synthesized via a bromide-induced process as described elsewhere.¹⁹ Figures 1a–1h present the powder XRD patterns and corresponding TEM images of as-synthesized Fe_7C_3 , Fe_2C , Fe_5C_2 , and $\alpha\text{-Fe}$ NPs. According to the XRD results, the peaks in each sample consistent with the standard patterns, which suggests the generation of a single phase in each of the four samples. Furthermore, TEM images of the four samples indicated that they were all spherical NPs with diameters around 18 nm.

The XAFS and Mössbauer spectra data also supported this conclusion. Fe K-edge XANES suggested that the iron-carbide particles exhibit very low oxidation states as compared with the metallic Fe foil. The relatively low-frequency oscillation of the postedge features indicated that the Fe central atom had neighbors with small bond lengths. Further extended XAFS (EXAFS) fitting results confirmed that all of the Fe-carbide particles synthesized had Fe–C coordination shells near 2.0 Å. The average first Fe–Fe shell bond length expanded from 2.46 to around 2.60 Å due to the incorporation of carbon into the BCC lattice of $\alpha\text{-Fe}$ (Figures 2a and 2b). Furthermore, no features of Fe oxides and Fe were observed, suggesting all the particles were pure carbide. (The Mössbauer spectra and corresponding fitting results are shown in Supporting Information Figure S4 and Table S1.) The sextet peaks indicated the formation of Fe_2C , Fe_7C_3 , and Fe_5C_2 NPs, while the weak doublet peaks in

Fe_5C_2 suggested that the Fe_5C_2 NPs might have better crystallinity compared with Fe_2C and Fe_7C_3 . Therefore, the single-phase nature of these $\alpha\text{-Fe}$ and iron-carbide NPs, along with their similar morphology, provides us with an ideal platform for the investigation of their intrinsic catalytic behavior and structural evolution in the FTS process. At the same time, because the cementite (Fe_3C) was reported to have poor activity in FTS,^{30,32} it was not discussed in this work.

The $\alpha\text{-Fe}$ and iron-carbide NPs were dispersed on silica support and directly used in the FTS reaction (3 MPa syngas, 270 °C). For a 40 h reaction, the product distribution on Fe and Fe_5C_2 was similar except Fe had higher C_{5+} selectivity (49%). Fe_2C had the highest selectivity toward CO_2 (22.7%), while Fe_2C and Fe_7C_3 showed considerably high selectivity of 20.1% and 19.5% toward methane, respectively (Supporting Information Figure S5). Both CO_2 and CH_4 are undesired products. CO conversion on prepared catalysts with time on stream is shown in Figure 3 with FTS activity shown in Table 1.

At the beginning, Fe_5C_2 exhibited the highest CO conversion (around 35%), followed by Fe_2C (29.5%) and Fe_7C_3 (22.5%), while $\alpha\text{-Fe}$ had the lowest CO conversion (15.2%). The initial CO reactivity reflected the intrinsic catalytic properties of Fe_2C , Fe_7C_3 , and Fe_5C_2 . With the reaction proceeding, the activities of the four catalysts showed different trends. For Fe_5C_2 and Fe_2C , the CO conversion decreased sharply in the first 6 h but

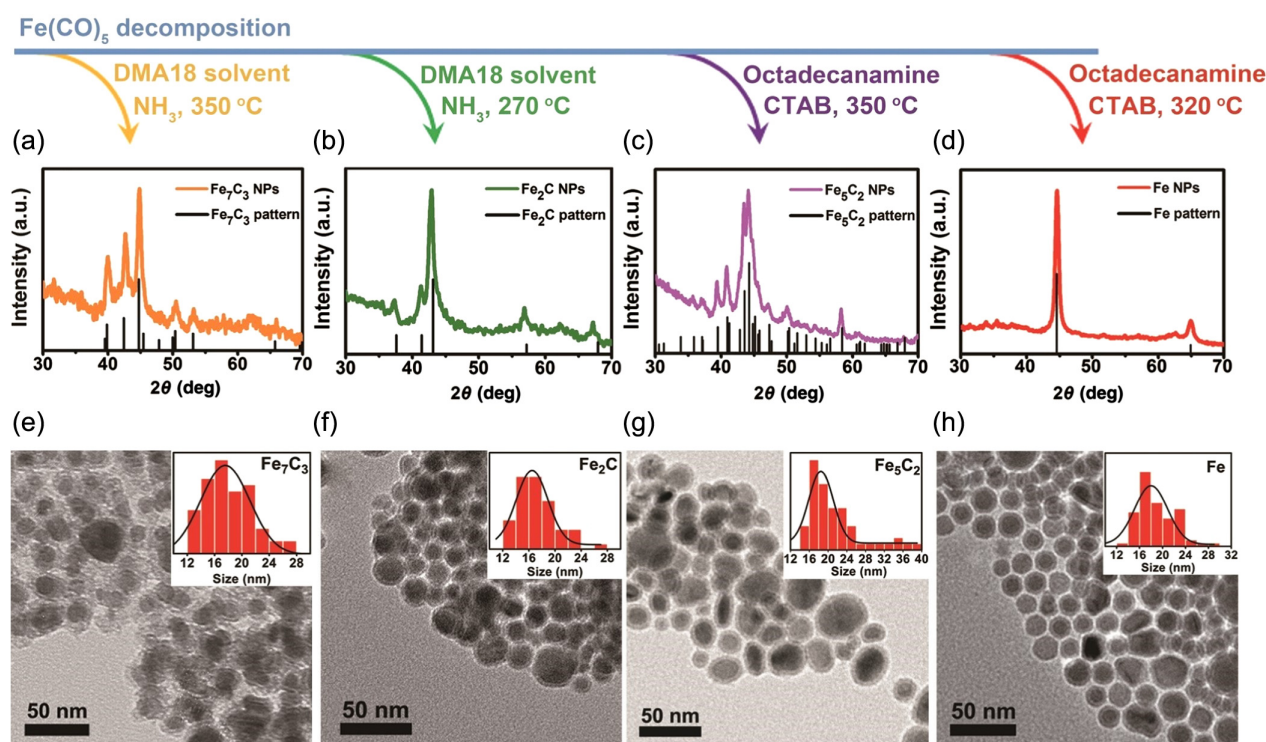


Figure 1 | XRD patterns and TEM images of (a and e) Fe_7C_3 , (b and f) Fe_2C , (c and g) Fe_5C_2 , and (d and h) $\alpha\text{-Fe}$ NPs. It is clear from the TEM images and XRD profiles that all the obtained Fe and iron-carbide NPs have the pure-phase structure and similar size (around 18 nm).

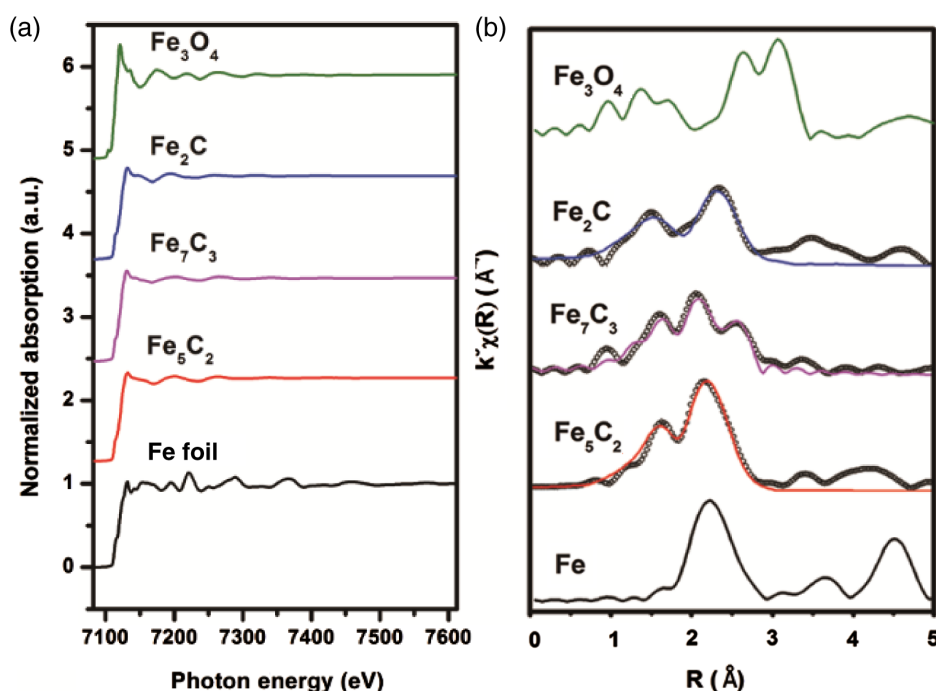


Figure 2 | (a) XANES and (b) EXAFS spectra of Fe_5C_2 , Fe_7C_3 , and Fe_2C . The XANES and EXAFS spectra of Fe foil and standard pattern of Fe_3O_4 are shown by black and green curves, respectively. The curves composed by black circles in (b) indicate the experimental data of each iron carbide, and the colored curves indicate the simulation curves. The XAFS samples were prepared and transferred under protection of Ar.

gradually stabilized later on. For Fe_7C_3 , CO conversion dropped slightly in the first 5 h, and after a small increase, it became stabilized. On the contrary, that of the α -Fe catalyst increased rapidly in the first 7 h, and then remained almost constant afterward. When the conversion for all catalysts approached the steady state, Fe_5C_2 remained the most active with a conversion of 27%, and the activities of Fe_7C_3 and Fe (25% and 24%) were slightly lower, whereas Fe_2C became the least active (17%). The turnover frequency (TOF) and activity of

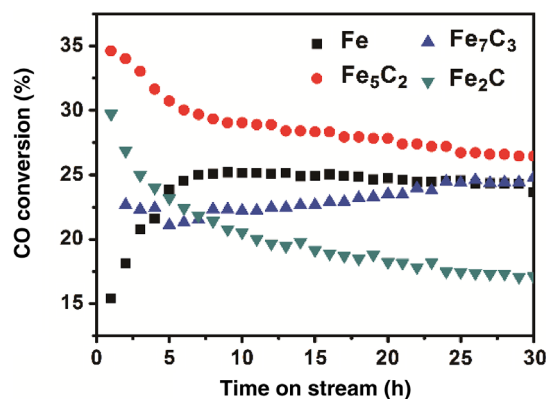


Figure 3 | CO conversion as a function of time over Fe, Fe_2C , Fe_7C_3 , and Fe_5C_2 catalysts (reaction conditions: 270 °C, 30 bar, 20 mL/min syngas).

those Fe-based catalysts are summarized in Table 1. Obviously, Fe_5C_2 had the highest initial- and steady-state activities (2.3 and $1.8 \times 10^{-4} \text{ mol}_{\text{CO}}/\text{g}_{\text{Fe}}\cdot\text{s}$). After reaction, the morphologies of used Fe_5C_2 and Fe_2C catalysts were maintained with main phases transformed into a mixture of iron oxide and original carbide, whereas for Fe_7C_3 , the formation of Fe_5C_2 was observed (Supporting Information Figures S6 and S7). Significantly, the main phase of used α -Fe catalyst was transformed into Fe_5C_2 , in addition to Fe_3O_4 (Supporting Information Figure S6). There has always been a debate on whether and why metal and/or carbide is the active phase for FTS. To examine this, we chose Fe_5C_2 as a representative for the carbide phase and compared its catalytic behavior in the early stage of FTS with that of α -Fe catalyst. We designed a high-pressure STPSR apparatus to allow the observation of reaction kinetics at high pressure. This STPSR experiment enabled us to observe the masked information by steady-state reaction evaluation.^{17,60,61}

The formation of various products during the STPSR of syngas over the Fe_5C_2 catalyst is illustrated in Figure 4a. Before 150 °C, no products were formed, which indicates that syngas cannot be activated over Fe_5C_2 at temperatures below 150 °C. With the reaction temperature reaching 150 °C for 20 min, the formation of water, C_2 hydrocarbons (mostly acetylene and ethylene), and CO_2 were observed. When the temperature was raised to 170 °C, with the appearance of C_3 hydrocarbon, the

Table 1 | The Surface-Specific Activity (TOF) and Metal-Mass-Based Activity (Activity) Excluded CO₂ Formation over Various Iron and Iron-Carbides Catalysts

Catalysts	Temperature (°C)	Pressure (bar)	TOS (h)	TOF _{FTS} (s ⁻¹)	Activity (1 × 10 ⁻⁴ Mol _{CO} /g _{Fe} ·s)	References
Fe	270	30	1	0.13	1.1	This work
Fe	270	30	30	0.20	1.7	This work
Fe ₅ C ₂	270	30	1	0.29	2.3	This work
Fe ₅ C ₂	270	30	30	0.22	1.8	This work
Fe ₇ C ₃	270	30	1	0.16	1.6	This work
Fe ₇ C ₃	270	30	30	0.18	1.7	This work
Fe ₂ C	270	30	1	0.16	1.8	This work
Fe ₂ C	270	30	30	0.09	1.0	This work
Fe _x O _y @C	270	20	—	—	0.31	[55]
Fe/SiO ₂	270	20	—	—	0.20	[56]
Fe-in-CNT	270	50	—	—	2.5	[57]
25-Fe@C	340	20	—	0.11	4.9	[58]
RQ Fe	200	30	—	—	3.5	[59]

Abbreviations: TOF, turnover frequency; TOS, time on stream.

amount of water and C₂ formed was considerably high. Yet, to our surprise, methane was not detected until then. The MS signal for methane appeared only when the reaction temperature reached 190 °C. The formation of methane at a higher temperature than that of C₂ hydrocarbons indicates that the hydrogenation of monomers toward methane is kinetically less favorable than C-C

coupling. When the temperature reached 210 °C, C₄ compounds appeared, followed by the formation of C₅₊ hydrocarbons 20 min later.

To our surprise, the STPSR profile of α-Fe catalyst showed a distinct behavior as compared with that of Fe₅C₂ (Figure 4b). At 150 °C, water formation was reproducibly observed, whereas no C₂ hydrocarbons could be detected in 2 h. Instead, C₂ hydrocarbon appeared only at around 10 min after the temperature reached 170 °C. The formation temperature of methane, C₃ alkane, and CO₂ was at least 190 °C. C₄ and C₅ formed at 210 and 230 °C, which was very similar to those of Fe₅C₂. Thus, although C₂ hydrocarbon on α-Fe formed at a higher temperature (170 °C) than that on Fe₅C₂ (150 °C), the formation of methane on both catalysts requires a temperature of 190 °C or higher. Namely, methane formation on both catalysts remains less favorable, compared with the formation of water and C₂ hydrocarbon. The formation of water in STPSR process is interesting as the preadsorbed water on the catalysts was removed through a pretreatment at 300 °C before STPSR. Therefore, the water can only be formed from the reaction between hydrogen and oxygen adsorbed on the catalyst. There are two possible sources for adsorbed oxygen, namely, either the product from CO dissociation or the contaminated molecular oxygen from the gas phase that was not removed by hydrogen pretreatment before STPSR. In fact, the latter oxygen source can be excluded by a H₂-STPSR experiment (Supporting Information Figure S8). This implies that the oxygen source of water on the α-Fe catalyst can only be the dissociated O from CO. Therefore, it is concluded that CO dissociation took place at 150 °C over Fe catalyst, which is also the case for the Fe₅C₂ catalyst.

Dissociated carbon would be hydrogenated, subsequently forming the hydrocarbons.^{62,63} At 150 °C, the

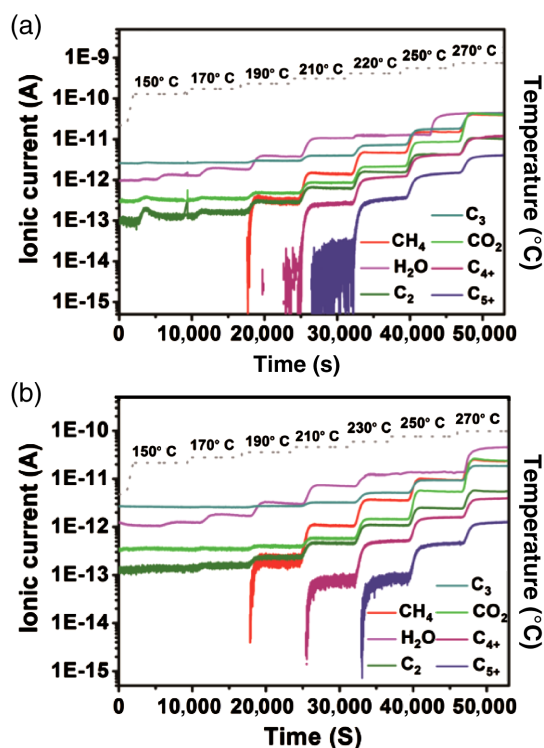


Figure 4 | STPSR profile on (a) Fe₅C₂ and (b) Fe catalysts.

formation of the hydrocarbons was observed on Fe_5C_2 , but not on $\alpha\text{-Fe}$ within the first 2 h. This implies that dissociated carbon would accumulate on $\alpha\text{-Fe}$, and might diffuse into the interstitial sites of the subsurface and the bulk $\alpha\text{-Fe}$ region, carburizing $\alpha\text{-Fe}$ toward carbide. Fe NPs can be easily carburized into iron carbides,⁶⁴ whereas the transformation from the single crystals and sheets of Fe to iron carbide is difficult,^{65,66} which can be attributed to the difficulty for carbon atoms permeating into the interstitial sites formed by the close packing of Fe atoms. Thermodynamically, this is possible since the corresponding Gibbs free formation energy (-3.22 eV per chemical formula unit of Fe_5C_2 under FTS conditions) at this condition is exothermic (Section 8 in [Supporting Information](#)). The above hypothesis was confirmed by in situ XRD experiments under FTS reaction conditions (2 MPa syngas, Figure 5). It was found that under a syngas stream, $\alpha\text{-Fe}$, which corresponds to a diffraction with a two-theta value of 35.3° , was gradually transformed to the iron-carbide phase (two-theta value of 34.8° , most like Fe_5C_2) when increasing the reaction temperature higher than 220°C . Although the reaction condition was not exactly the same as in the STPSR experiments, this tells compellingly that in the FTS process, $\alpha\text{-Fe}$ is indeed apt to react with surface carbon from dissociated CO, forming iron carbide. Moreover, when we carburized the supported $\alpha\text{-Fe}$ catalyst with ethylene or the mixture of ethylene and hydrogen to get supported iron-carbide catalyst (Fe_5C_2 , as evidenced by XRD (Figures 6b and 6c), we observed that the initial activity of the resulting carburized catalyst increased dramatically (Figure 6a), reaching 33%, very close to the initial

activity of the pure-phase Fe_5C_2 catalyst. The activity evolution of the carburized Fe catalyst almost duplicated that of Fe_5C_2 catalyst, that is, it dropped gradually in the first few hours and became relatively stable after around 15 h of reaction. The induction period observed on the $\alpha\text{-Fe}$ catalyst could therefore be attributed to the process of carburization. Once the carburization was complete and Fe_5C_2 was formed, the resulting catalysts would show a higher activity because of the higher intrinsic activity of Fe_5C_2 .

For the drop of activity of Fe_5C_2 catalyst with reaction time (Figure 3), it was attributed to the oxidation of iron carbide by the oxidative products such as H_2O and CO_2 to the inactive iron oxide. This was confirmed by the activity measurement of pure-phase Fe_5C_2 treated with CO_2 at different temperatures (see [Supporting Information Figure S9](#)). Indeed, both the initial crystal phase of the catalyst and the reaction atmosphere could affect the structural evolution and eventually the equilibrium structure of iron-based catalysts under reaction conditions.

To rationalize the experimental results presented above, it is essential to study the intrinsic FTS activity of the pristine iron and the difference with the iron

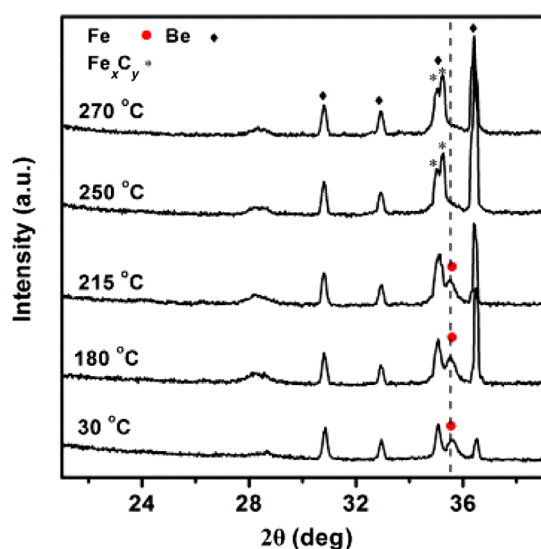


Figure 5 | In situ XRD patterns of Fe NPs treated with 2 MPa syngas at various temperatures. (XRD was recorded in beam line 14B of SSRF. The diamonds mark the diffractions from beryllium-made in situ cell.)

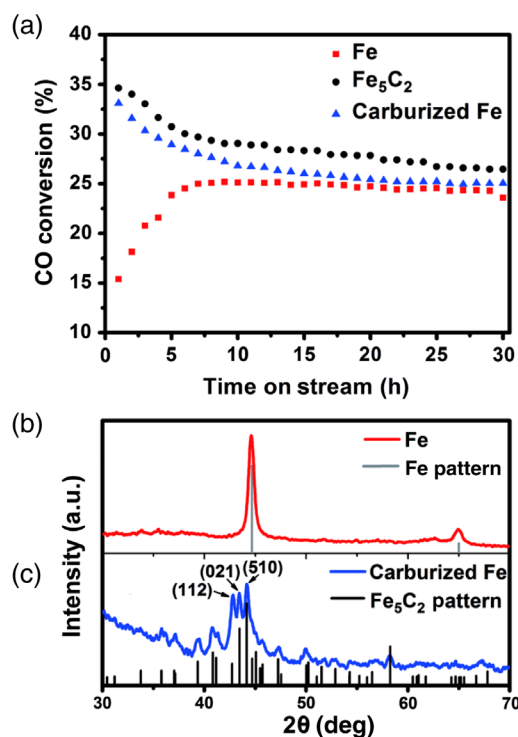


Figure 6 | (a) CO conversion of $\alpha\text{-Fe}$, Fe_5C_2 , and pre-carburized $\alpha\text{-Fe}$ NPs catalysts. XRD pattern of $\alpha\text{-Fe}$ NPs catalysts (b) before and (c) after carburization. It is clear that carburizing the supported $\alpha\text{-Fe}$ catalyst with ethylene or the mixture of ethylene and hydrogen at 350°C for 1 h would transform $\alpha\text{-Fe}$ to Fe_5C_2 .

carbide. Herein, DFT calculations were performed to study the crucial FTS steps including CO activation, C-C coupling and methane formation. The stepped Fe (310) and Fe-terminated Fe_5C_2 (100) surfaces (Supporting Information Figure S10) were used to model the corresponding iron and iron-carbides catalysts, respectively. The stepped Fe (310) was chosen here because the Fe (310) surface occupies a large proportion of surface area of the iron Wulff shape by 22%⁶⁷ and it shows higher activity for CO dissociation as compared with the traditional (110), (100), (211), and (111) surfaces.⁶⁸ Here, Fe_5C_2 (100) was selected since it is one of the largest exposed surfaces under operating FTS conditions (600 K, 10 bar, $\text{H}_2/\text{CO} = 2.5$) on Fe_5C_2 Wulff construction.⁶⁹ On a C-terminated Fe_5C_2 surface, there will be no active sites for CO activation or the activity of CO activation will be significant low.^{70–72} Herein, we have adopted Fe-terminated Fe_5C_2 (100) surface which stands for the C-terminated Fe_5C_2 (100) surface with abundant C vacancies, where CO dissociation is feasible.^{39,70} In addition, Fe (310) and Fe-terminated Fe_5C_2 (100) surfaces have the same B5 step surface, and it is feasible for the direct activity and selectivity comparison between Fe and Fe_5C_2 .^{73,74} The calculated binding energetics of the important intermediates CO^* , C^* , and CH^* at their most favorable sites were -2.07 , -8.15 , and -6.90 eV on Fe (310), while -1.91 , -7.08 , and -6.53 eV on Fe_5C_2 (100) (Supporting Information Table S2), respectively. As expected, pristine iron is more reactive than iron carbide, and the binding strength toward C^* is 1.07 eV stronger. This is determinative to the distinct activities of metal and carbide phases on CO activation, C-C coupling, and methane formation.

For Fe (310), calculated E_{app} for CO dissociation was 0.93 eV, with an exothermic reaction energy (-0.98 eV) due to the strong C-Fe binding (Figure 7a). The small

barrier is in good accordance with observation of water formation at 150 °C. For C-C coupling, there are many possible pathways between CH_i and CH_j ($i, j = 0-3$). To evaluate their relative activity, equilibrium concentration of the corresponding monomers, which is proportional to their formation energy E_f (with zero-energy reference of CO dissociation product), should be considered. We define accordingly apparent barrier E_{app} , namely, summation of formation energy E_f of CH_i and CH_j and their coupling barrier E_c . It is found that the strong C-Fe binding not only makes the formation of CH_i and CH_j monomers energetically highly costly with E_f falling in a magnitude of 2.49 eV, but also makes CH_i - CH_j coupling kinetically very demanding with a maximum E_c of 1.99 eV (Table 2 and Supporting Information Figure S15). Indeed, among all possible CH_i - CH_j coupling considered, the least E_{app} ($= E_f + E_c$) calculated was as large as 1.87 eV from C^* - CH^* coupling. Methane formation was also found to be highly demanding with an overall barrier of 2.27 eV (Supporting Information Table S3 and Figure 8b).

The large barriers for C-C coupling and methane formation on Fe (310) lead to a rather poor FTS activity for the pristine iron. Our DFT calculations are also consistent with previous experimental results that iron catalysts containing no carbide have the lowest FTS activity.²⁸ The dissociated carbon from CO would therefore be sufficient for a long time on the catalyst surface, which is necessary for carburization and transition toward more thermodynamically favorable carbide. As a result, the unsaturated C_2 hydrocarbon observed on iron at 170 °C (Figure 4b) cannot come from the pristine iron. Instead, it comes from the carburized iron catalysts, namely, iron carbide. Since dissociated carbon was already available at 150 °C on the pristine iron, this also means the carburization process requires the activation temperature of at least 170 °C.

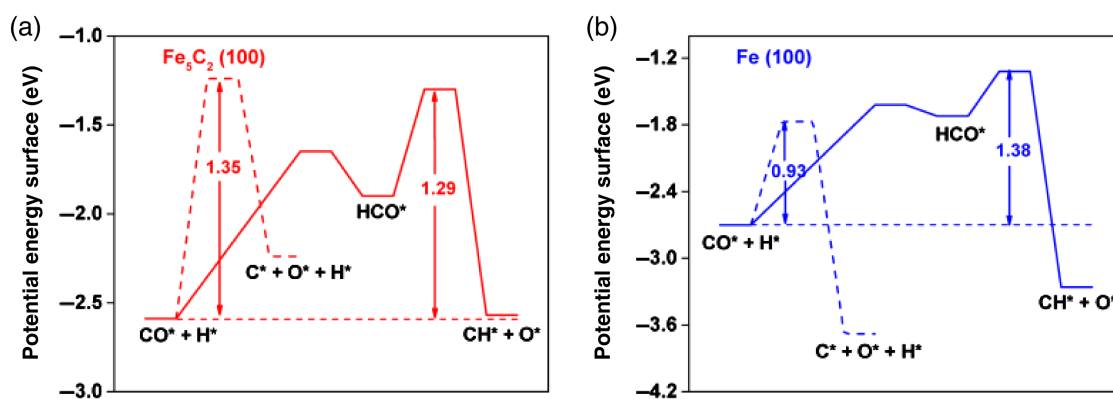


Figure 7 | The potential energy diagrams for CO dissociation on (a) Fe_5C_2 (100) (red) and (b) Fe (310) (blue) surfaces. The solid and dashed lines present direct and H-assisted CO activation pathways, respectively. The apparent activation barriers (in eV) are indicated. The corresponding geometries involved in CO activation are shown in Supporting Information Figures S11 and S12.

Table 2 | Calculated Energetic for $\text{CH}_x\text{-CH}_y$ Coupling over $\text{Fe}_5\text{C}_2(100)$ and $\text{Fe}(310)$ Surfaces. E_f , E_c and E_{app} are the Formation Energy of CH_x^* and CH_y^* , Elementary Coupling Barrier between CH_x^* and CH_y^* , and the Apparent Barrier on $\text{Fe}_5\text{C}_2(100)$ and $\text{Fe}(310)$ Surfaces. The Formation Energy Was Calculated with Respect to CH^* on $\text{Fe}_5\text{C}_2(100)$ Surface and the Atomic Carbon C^* on the $\text{Fe}(310)$ Surface Since They Are Located at the Lowest Valley of the Whole Potential Surface

Reaction	E_c (eV)	$\text{Fe}_5\text{C}_2(100)$ E_f (eV)	E_{app} (eV)	E_c (eV)	$\text{Fe}(310)$ E_f (eV)	E_{app} (eV)
C^*-C^*	0.84	0.47	1.31	1.99	0.00	1.99
C^*-CH^*	0.78	0.24	1.02	1.45	0.42	1.87
C^*-CH_2^*	0.77	0.59	1.36	1.16	1.20	2.36
C^*-CH_3^*	1.13	0.52	1.65	1.03	1.29	2.32
CH^*-CH^*	1.02	0.00	1.02	1.38	0.85	2.23
$\text{CH}^*-\text{CH}_2^*$	0.60	0.36	0.96	1.24	1.63	2.87
$\text{CH}^*-\text{CH}_3^*$	1.19	0.28	1.47	1.35	1.71	3.06
$\text{CH}_2^*-\text{CH}_2^*$	0.48	0.72	1.20	0.79	2.41	3.20
$\text{CH}_2^*-\text{CH}_3^*$	1.16	0.64	1.80	1.07	2.49	3.56

For $\text{Fe}_5\text{C}_2(100)$, the calculated E_{app} for CO dissociation is 1.29 eV with an endothermic reaction energy (Figure 7b), due to the destabilization of dissociated C^* on the carbide surface. Though the CO dissociation barrier is higher than that of $\text{Fe}(310)$, its modest value makes CO dissociation on $\text{Fe}_5\text{C}_2(100)$ remain facile, for instance at 150 °C. This is still in accordance with experimental observation of water formation at this temperature. On the other hand, the destabilization of surface C^* and CH^* on carbide (Supporting Information Table S2) greatly promotes the C-C coupling and methane formation. The formation of CH_i and CH_j monomers becomes energetically much less costly with E_f falling in the range of 0–0.72 eV, and the $\text{CH}_i\text{-CH}_j$ coupling becoming kinetically facile with a maximum E_c of 1.19 eV (Table 2 and Supporting Information Figure S16). As a result, the least

E_{app} calculated for the C-C coupling is only 0.96 eV from $\text{CH}^*-\text{CH}_2^*$ coupling. Moreover, C^*-CH^* , CH^*-CH^* , and $\text{CH}_2^*-\text{CH}_2^*$ coupling are also kinetically favorable with E_{app} of 1.02, 1.02, and 1.20 eV, respectively. For methane formation, the overall barrier also decreases significantly to 1.38 eV (Supporting Information Table S4 and Figure 8a).

As can be seen in the reaction barriers in Figure 9, C-C coupling barriers are even lower as compared with CO activation on $\text{Fe}_5\text{C}_2(100)$ with a barrier of 1.29 eV. This means once CO dissociates on Fe_5C_2 , all these C-C coupling pathways could take place right away. This nicely corroborates the experiment with Fe_5C_2 at 150 °C (Figure 4a), namely, once syngas was activated, both acetylene and ethylene were observed simultaneously with the formation of water. Meanwhile, the barrier for

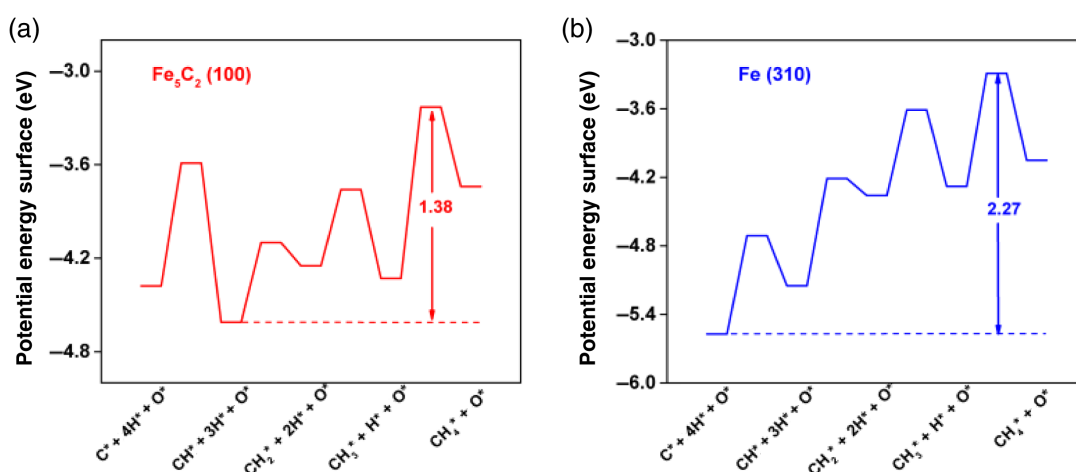


Figure 8 | The potential energy diagram for methane formation on (a) $\text{Fe}_5\text{C}_2(100)$ (red) and (b) $\text{Fe}(310)$ (blue) surfaces. The apparent activation barriers (in eV) are indicated. The corresponding configurations for intermediates adsorption and transition states are shown in Supporting Information Figure S13 and Figure S14, respectively.

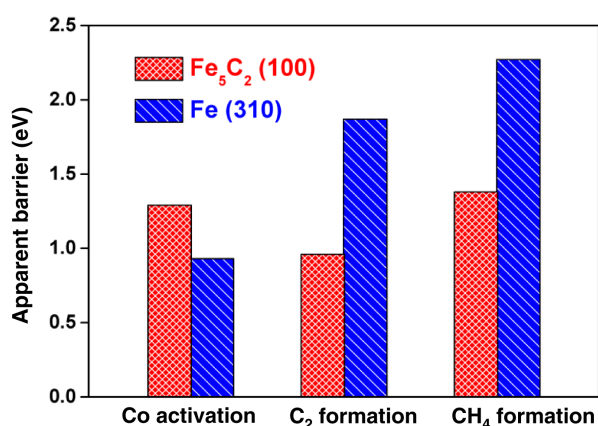


Figure 9 | Apparent barriers comparison for CO activation, C₂, and CH₄ formation on Fe₅C₂ (100) and Fe (310) surfaces.

methane formation remains higher than that of CO activation, a fact that was also the case with Fe (310). This implies that a higher temperature for methane formation would be required, which was indeed found in the corresponding STPSR experiment (Figure 4a).

Conclusion

Through a modified liquid-phase route, a series of pure-phase metallic iron and iron-carbide NPs were successfully synthesized, providing an ideal platform to investigate the fundamentals in an iron-based FTS reaction that has not previously been revealed. Fe₂C, Fe₇C₃, and Fe₅C₂ were all found to be active for FTS reaction, with Fe₅C₂ the most active phase. By using a house-designed STPSR method, transient information about the surface species and its reactivity was revealed. DFT calculations showed that iron carbide is intrinsically more active than the pristine iron for C–C coupling and methane formation because of the strong binding of dissociated atomic carbon on iron. Furthermore, C–C coupling was easier compared with methane formation on iron carbide, which makes iron carbide highly active for FTS with good olefin selectivity. The distinct activities of different iron phases (metal vs carbide) also revealed their stability under FTS conditions could be used to design more efficient iron-based FTS catalysts.

Additions and Corrections

a. Page 2712, The author list was incomplete in the original paper. Two names marked as bold below have to be added as the coauthors who did the Mössbauer Spectroscopy characterization.

“Huabo Zhao^{1†}, Jin-Xun Liu^{2†}, Ce Yang^{3†}, Siyu Yao¹, Hai-Yan Su⁴, Zirui Gao¹, Mei Dong⁵, **Junhu Wang⁶**,

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b. Page 2721, The Acknowledgment Section in the original paper was missing, and should have appeared as shown below:

Acknowledgment:

Work supported by Mössbauer Effect Data Center, Dalian Institute of Chemical Physics, Chinese Academy of Sciences, China.

These changes were made post publication on Mar. 8, 2021. All authors on the article are aware of and agree with the changes.

Supporting Information

Supporting Information is available and includes the XRD spectra (Figures S1 and S2); physical absorption characterization of the support material (Figure S3); Mössbauer effect spectra (Table S1 and Figure S4); products distribution of catalysts (Figure S5); TEM images and XRD patterns of used catalysts (Figures S6 and S7); TPSR of Fe and Fe₅C₂ in pure H₂ (Figure S8); XAFS spectra (Figure S9) and FTS performances of CO₂ treated Fe₅C₂ catalyst; energetic and geometric information for CO activation, CH₄ formation, and CH_x–CH_x coupling by DFT calculations (Tables S2–S4 and Figures S10–S16).

Conflict of Interest

There is no conflict of interest to report.

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